DEVELOPING CURATRACK: A DIGITAL HEALTH RECORDS MANAGEMENT SYSTEM

BY

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DATE SUBMITTED: 20TH OCTOBER 2025.

DECLARATION

I, Sally Njeri Kinyanjui, declare that this or any other university has not previously submitted this work for the awarding of the course marks. I also declare that I will only present this project once the supervisor has approved and signed the document. To the best of my knowledge and belief, this work contains no material previously published or written by another person except where due reference is made.

Name: Sally Njeri Kinyanjui

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APPROVAL

This project proposal of Sally Njeri Kinyanjui was conducted under our supervision and is submitted with our approval as university supervisors.

Supervisor Name: …….. Signature: ……

Date: ……

DEDICATION

I take this opportunity to dedicate this piece of work to my loved ones and my country, Kenya, which I love so much.I put in a lot of effort to ensure the success of my project so it can benefit the people of my country.

ACKNOWLEDGEMENT

I sincerely thank my coursemates and family members for the support they gave me while I was working on this project. My course mates helped me in areas where I was stuck while working on the project, and as for my family, I am grateful for the material and emotional support.

ABSTRACT

An evident mismatch in the old system, which uses paper, has brought to the fore the growing need to ensure that medical records are managed efficiently, safely, and conveniently. It turns out, however, that hospitals and clinics have very serious issues such as the loss of data and duplicate responses, which predisposes great difficulty in obtaining patient information in a short time. That basically leaves a dent in the way patients are treated and the decision-making of doctors. It is the reason why we have to develop digital health solutions, which are precise, swift, confidential, and simple for accessing patient information.

Curatrack is an intelligent health-management platform. It is based on a web interface written on a database architecture, which is capable of updating and monitoring the healthcare status of patients in real time. Curatrack aids in increasing patient-provider interaction through records made available to each party and enhancing privacy as well as making the entire clinic run smoothly. It also streamlines the back-and-forth of the providers and patients, hands them ready records, improves security, and generally improves service delivery. In testing we observed the extent to which we found the record management accurate and user-friendly as well as the extent to which Curatrack proved to be effective. The fact that the system is automated saved hours of work and ensured that data was not duplicated in different modules. Curatrack is open and a combination of tech and health management to empower doctors to make even smarter choices.

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D.F.D—Data Flow Diagram

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1.1 Background Information

Healthcare is a crucial factor in enhancing the quality of life. Surprisingly, patient information still remains a significant challenge in low-income nations. Primarily, the smaller health facilities store their data in paper form thereby leading to repetitive tasks, the chance of data loss or corruption, and even fraud. These issues lead to service delays, patient care threats, and submissiveness of hospital management.

Technology is changing fast, hence an actual opportunity to digitize healthcare processes in an easy-going and cost-effective manner. That is why we should have such a system as CuraTrack, an electronic health record solution for smaller clinics. The title is also rather intelligent: "cura" in Latin means "care," and "track" implies tracking progress. Its primary target would be the small and medium-sized facilities, and the main objective is to become the location centered around patient data. The problem of record-keeping, which is rather time-consuming, may be overcome with CuraTrack implementation: improving patient outcomes and clinic workflow, reducing paperwork, accelerating services, and increasing their efficiency can be of great benefit to both patients and providers.

1.2 Problem Statement

In most of the facilities, healthcare providers have difficulties in keeping patient records that are accurate, accessible, and secure. Physical systems are susceptible to loss, duplication, and damage, and those available as digital alternatives are expensive or complicated to support smaller clinics. Consequently, it will be time-consuming to access patient histories, the risk of misdiagnosis may occur since the information will be incomplete, and follow-up care among the long-term patients will become ineffective. These inconveniences lead to a lack of patient satisfaction and poor service delivery. There is thus the necessity of a cost-effective, secure, and easy-to-use digital health records management system that provides greater accessibility, data confidentiality, and continuity of care.

1.3 Objectives

General Objectives

Bring CuraTrack to life, a computerized system for recording and keeping patient information while also improving healthcare

delivery.

Specific Objectives

1. To provide a centralized system for storing and retrieving patient records in a secure database.
2. Introducing modules for patient registration, appointment scheduling, and follow-up tracking.
3. To assist in the documentation of medical history: allergies/conditions.
4. Produce clinical and administrative reports to aid decision-making.
5. Integrate access control and authentication restrictions to protect patient privacy and security.

1.4 Justification

Management of health records needs to be done in an efficient way in order to provide quality healthcare services. CuraTrack replaces the problem of manual record-keeping and unavailable patient history with a system that is easy to use, scalable, and relatively inexpensive.

To the healthcare providers, the system will save time that is spent on administrative work and errors. To patients, it will enhance continuity of care by having the right and available records. To administrators, it will give them a platform to make decisions based on data using reporting capabilities.

Furthermore, the system can also be expanded to work with larger health information networks in the future, and it will be applicable at the clinic level as well as in the broader healthcare systems.

1.5 Scope

It will be directed at developing the prototype of CuraTrack that may be sold to small and medium-sized clinics. The functional aspects that will be covered are

- Patient Management. Basic demographic information, such as medical history, allergies, and underlying disorders, is captured and stored.
- Doctors and nurses can make follow-up appointments.
- Medical records contain information about the diagnosis, treatment, therapy, and recommended drugs.
- Reporting: Create simple reports such as patient lists, appointment schedules, and medical histories.

- Security: Users must be checked before being provided access to secret data.

At this stage, the system will not be linked to more advanced services like national health databases, billing systems, or laboratory information systems. However, the design will be customizable in the future to accommodate such functions.

1.6 Relationship to my Diploma Program

This project directly relates to my Diploma in Computer Science course, as it involves identifying a real-world problem and offering a solution. Computer science is a broad field that focuses on identifying real-world problems and using various resources and technologies to develop solutions and recommendations.

CuraTrack, therefore, relates to my course because, through the use of various technologies, it solves problems faced inhealthcare facilities. In this Computer Science Diploma, we studied database integration using SQL. In the future, knowledge can be used, and the project can be integrated with national records, making it useful at the national level and in bigger facilities.

CHAPTER 2

2.1 Introduction.

Almost every business, and especially healthcare, is subject to change due to the changing of the world and the changes prevailing in technology. In the past, medical institutions used manual communication. Patient data would be documented and stored by the nurses and other healthcare staff to be used later. This is not as easy as it may seem, as it had its challenges. Some of the issues included duplication, loss of patient information, data mishandling, and unauthorized access by people. EHRs have been created in order to overcome these concerns.

Electronic health record (EHR) refers to the health data that is electronically maintained in the course of the lifetime of an individual in a bid to foster continuity of care, learning, and research. (Ajami et al. 2013). EHRs have led to uniformity in data collection and facilitated the retrieval of patient data for clinical care and research (Slika et al., 2005). Some of the main benefits of EHR systems that have been identified include reducing medical errors, improving quality of care,

conserving physicians' time, sharing patient information among healthcare practitioners, and workflow efficiency (Silka et al., 2005).

Despite EHRs being of much importance to the healthcare system, not all healthcare facilities have been able to implement them due to some constraints. Studies conducted have identified that the major barrier to adoption of EHR systems is a misalignment of cost and funds or financial reimbursement. Other obstacles include EHRs are more difficult to use than paper-based records and risks of privacy and security (Ajami et al., 2013) There are more problems, but for this project we will focus on the ones listed above.

The current chapter will seek to revise the literature available, such as the publications of the industry and the concept of technologies behind CuraTrack, a simple digital health records management system, to establish a starting point for the project and explain why secure, cost-effective, and efficient electronic health records can be developed and adapted to limited-resource clinics. The review is informed by the existing sources, such as Google Scholar and WHO reports, and is separated by the topics into three large parts:

History of health records management.

EHR database systems and EHR designs.

DHS security.

This framework can help me to recognize concepts, gaps, and how CuraTrack can fit. Being a computer science diploma student, I am keen on applying this in real life with such skills I have gained in databases and programming in Python. I will develop a Python SQL database with MySQL to create a simple system of patient storage and retrieval that is easy to maintain and run in small clinics.

2.2 Body

2.2.1 History of Health Records Management.

Health record management has become digital with its own set of challenges. Paper-based

systems were cheap, though they were also subject to mistakes, as Kruse et al. (2018) found that an error was caused by up to 20% of the mistakes, such as medical drugs of the wrong dosage being dispensed because files were lost, and they might take time to retrieve the files. Only 20% of clinics in other low-income countries use digital technologies because the costs can be prohibitively high and implementation is limited by inadequate infrastructure (WHO, 2020).

Paton et al. (2018) note that Kenya lags much even in comparison with the United States in adoption of EHR to conduct follow-ups with chronic-patient groups despite the fact that the U.S. had a short-term victory in terms of EHR roll-outs in the early 2000s.

The primary distinction that is pointed out is that small clinics require a more personalized and direct approach, but large hospitals rely on a saturated digital infrastructure. That is why CuraTrack is strategizing to be offline-friendly—to avoid the same traps that brought to struggle in the past.

2.2.2 Database Systems and Designs in EHR.

Sparsity of digital records also remains common in many low-income countries. WHO (2020) revealed that merely one out of five clinics utilizes a digital health technology, mostly due to it being prohibitive and due to the lack of a good infrastructure. Paton et al. (2018) report that Kenya has struggled to follow up with chronic patients, although success rates in the United States have improved since early EHR legislation in the 2000s. As a student, I have observed that digital technology is usually embraced by major hospitals that seek easy and affordable solutions, and small clinics are usually at the back of the technological implementation.

The CuraTrack was designed with the idea of filling this gap by working offline and simplifying it in minimal conditions. Patient data and appointments are stored in databases such as MySQL, which can be accessed immediately with the help of SQL queries (Chen et al., 2021). The open-source systems like OpenMRS are cheap and easily customized to suit local requirements (Fraser et al., 2010). Proprietary software, such as Epic, on the one hand, is expensive and demands professional IT services (Kruse et al., 2018). Digital technology enhances precision and efficiency, yet the small clinics continue to grapple with low accessibility to the internet and high prices. CuraTrack will seek to offer a lightweight system to enable the rapid registration of patients and the process of directing them to appointments.

Table 2.1: Comparison of EHR Technologies in a snap.

Type	Pros	Why it fts Curatrack
MySQL + Python	Good Performance and Free	Simple to use
Open - source	Customizable and no large expenditures	By implementing Python, the bookings and reports will become easy.

2.2.3 Digital Health Systems: Access and Security.

Any healthcare information system must involve data protection as a vital element. Fernandez-Aleman et al. (2013), Role-Based Access Control (RBAC) limits access to specific aspects of a patient's record, such as comprehensive data for clinicians and reports for administrators. CuraTrack integrates simple security solutions, such as Flask-Login, that are compatible with MySQL's permission system. Maintaining security in impoverished countries can be costly; thus, economical and safe solutions are critical (Oluoch et al., 2019; WHO, 2020). Open-source systems make it simple for employees to learn basic plugins without the need for advanced technical skills (Fraser et al., 2018).

CuraTrack's security features stress simplicity, accessibility, and data protection, allowing small clinics to safely manage sensitive health records.

2.3 Conclusion

This evaluation looks at how EHRs have improved record accuracy and speed around the world (Kruse et al., 2018; WHO, 2020). Smaller facilities, notably in Kenya, continue to encounter cost and infrastructure challenges (Panton et al., 2018). Affordable tools like MySQL and Python (Chen et al., 2021; Fraser et al., 2010), as well as security principles like RBAC (Fernandez-Aleman et al., 2013), can provide a comprehensive solution. CuraTrack offers a practical solution to implementing digital record-keeping in small clinics. It integrates security features for patient tracking, appointment management, and

reporting, bridging the gap between large hospital systems and underprivileged health centers.

CHAPTER THREE

3.0 Introduction

This chapter basically focuses on the tools, methods, and steps taken to bring our Electronic Digital Health Records System to work.

3.1 System Development Methodology

System development methodologies are structured approaches used to plan, manage, and execute the process of building software systems. Various methods can be used in different projects, but This is best suited by the prototyping model, the reason being that it provides constant feedback, flexibility, and user-centered development. Doctors, administrators, and receptionists are the users of this project whose requirements are different and dynamic, and might not be obtained in full at the initial stages. Through the prototyping technique, the developers can develop the first working system (prototype) within a short period of time to showcase the key functionality of the system, such as patient record management and appointments. It is a prototype that assists the users to visualize the system, as well as give them a chance to give feedback, as well as make suggestions on how the system can be improved during the early stages of the development process. This leads to the end system being more precise, easy to use, and more consistent with workflows in the healthcare environment. Also, the prototyping model minimizes the risk of expensive changes in the later stages of the development process, enhancing interaction between

The users and the developers, and make sure that the final product is successfully able to address the needs of the users.

3.2 Requirements Gathering

In this stage, the user and stakeholder needs are collected and analyzed to make sure that the system can address the real-life expectations and address the targeted problems. The idea is to learn what characteristics and functions are required to develop a secure, efficient, and easy-to-use system.

3.2.1 Collection Method

Questionnaires: This is used to gather data on large numbers of users by using written questions to aid in determining the requirements and the perceptions held on the system.

Surveys: A survey is a collection of general feedback or statistics within a group to know the general trends or likes.

Interviews: Entail face-to-face discussions with users to obtain in-depth information and precise needs.

Observation: Involves observing the way the users carry out tasks in order to know the

difficulties and to design the system better.

3.2.2 Non-Functional Requirements

- Security: The system should include user authentication and role-based access to safeguard all patient information.

- Usability: The interface should be user-friendly and easy for non-technical staff to operate with minimal training.
- Cost-Effectiveness: The system should rely on open-source tools and lightweight resources to remain affordable for small clinics.
- Reliability: Data must be stored accurately and consistently, with measures to prevent errors or information loss.

- Scalability: The design should support future upgrades, such as integration with national health data systems.

- Maintainability: The system should be simple to modify, troubleshoot, and update due to its modular structure.

3.2 Functional Requirements

Patient Management: Capture and store patient demographic details, medical history, and existing health conditions.

Data Access: Provide secure and centralized access to patient records for authorized users.

Appointments/Follow-Up: Handle scheduling of patient appointments and send follow-up reminders.

Medical Records: Maintain and update clinical information, including diagnoses and treatment details.

Reporting: Generate summaries such as patient lists, appointment overviews, and printable medical reports.

Security/Administration: Implement user authentication and role-based permissions to ensure data protection.

3.3 System Design

This section presents the architectural and logical representation of the CuraTrack system. **3.3.1 Block Diagram**

● The block diagram provides a high-level overview of the system's main modules and

how they interact. It illustrates the structure of the Medical Health Records Management System and the communication flow between components.

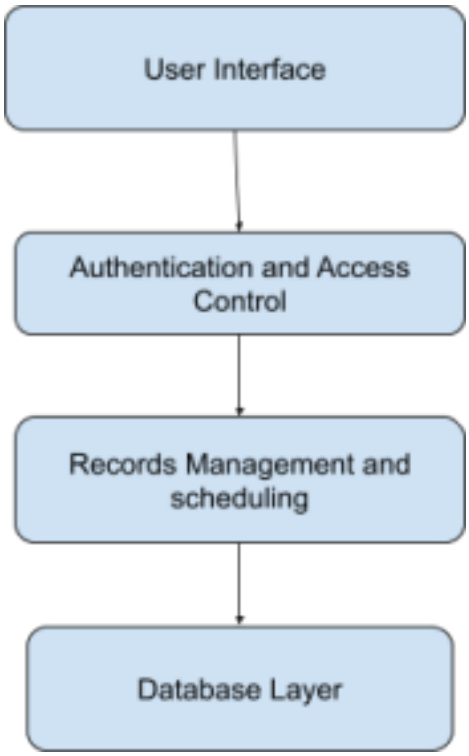


Diagram 3.1 CuraTrack Block Diagram

3.3.2 Data Flow Diagram (DFD)

The data flow diagram shows how information moves through the system, from user input to data processing and storage

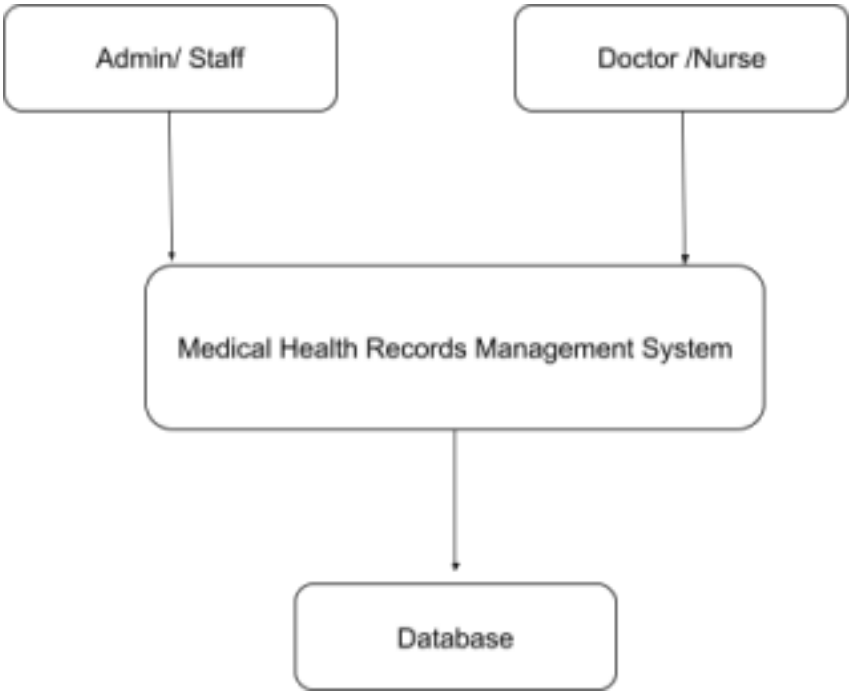


Diagram 3.2: Data Flow Diagram

3.3.3 Entity Relationship Diagram (ERD)

The ERD illustrates the data structure and relationships among entities: **Entity Key Attributes Relationships**

Patient PatientID, Name, DOB, Contact, Gender 1 Patient → Many Medical Records

Staff (User): UserID, Name, Role, Password 1 Staff → Many Appointments, Records

Appointment AppointmentID, Date, Time, Status Linked to Patient & Staff

Medical Record RecordID, Diagnosis, TreatmentPlan Linked to Patient & Staff

Medication MedName, Dosage Linked to Medical Record ER Diagram:

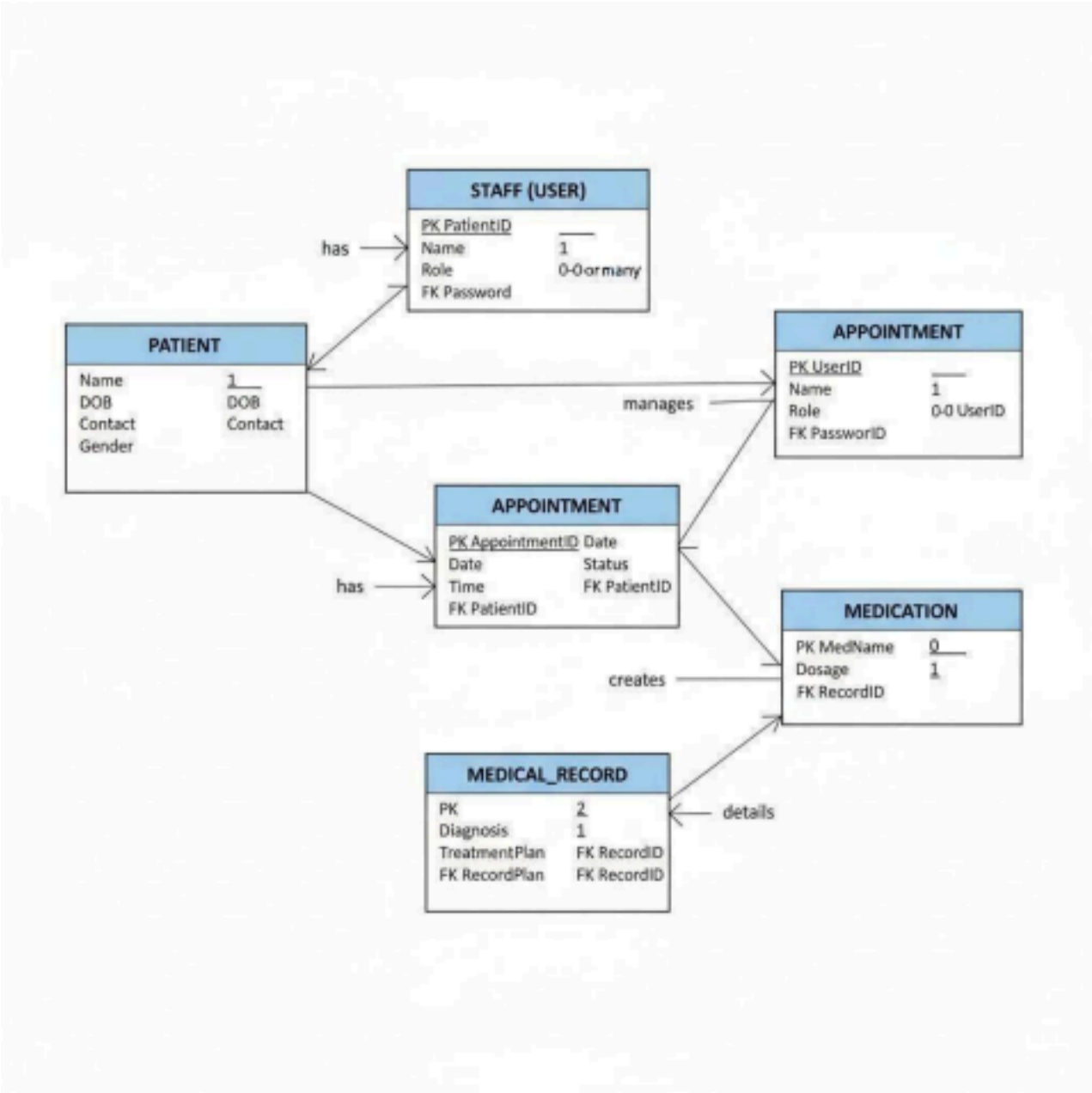


Diagram 3.3: ER Diagram

3.3.4 Use Case Diagram

A use case diagram represents the interaction between users (actors) and the system. It helps identify the different functions the system must perform from a user's perspective. In Cura Track, the main actors include

the admin, doctor, and receptionist. Each actor interacts with specific use cases such as managing patient records, scheduling appointments, and accessing reports. This diagram helps visualize system functionality and user roles clearly.

Actors:

- **Administrator:** Full access to manage users, system configurations, and reports.
- **Provider (Doctor/Nurse):** Access to patient records, diagnosis, and treatment entry.
- **Reception/Clerk:** Access to patient registration and appointment booking.

Core Use Cases:

Register Patient, Book Appointment, Record Medical Entry, Generate Reports, Manage Users, and Log Out. *Use Case diagram:*

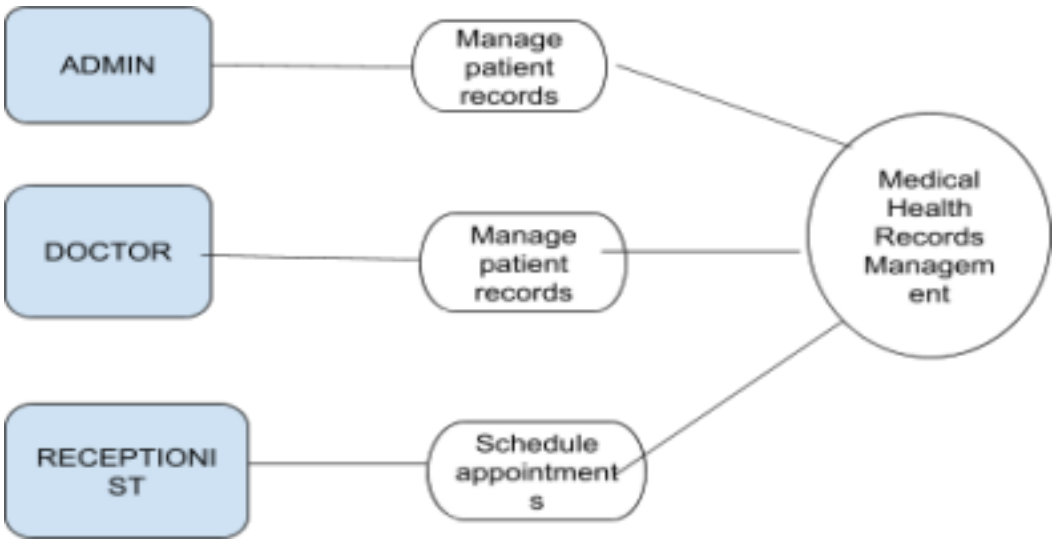


Diagram 3.4: Use Case Diagram

3.5 Technical Requirements

Category	Requirement	Rationale
Programming Language	Python	Clean object-oriented programming support and rapid prototyping.
	Open-source and enhances strong data	

Database MySQL

integrity.

Frontend HTML Enables a responsive, user-friendly web interface.

Development Tools VS Code Used for coding, version control, and system design.

Hardware (End-User) Laptop/PC, Tablet, Smartphone Ensures accessibility from multiple devices.

Software leading to easy access.

(End-User) 3.4 Budget

Web Browser eliminates installation needs.

Our project is quite affordable since with a budget of 70,000, we can actualize our project. The money is distributed so that 55,000 Kenyan shillings will be used to purchase a laptop for the project, and 10,000 Kenyan shillings will be used to pay for internet and hosting. The remaining 5000 can be used to pay for access to some applications.

Table 3.1 Budget table

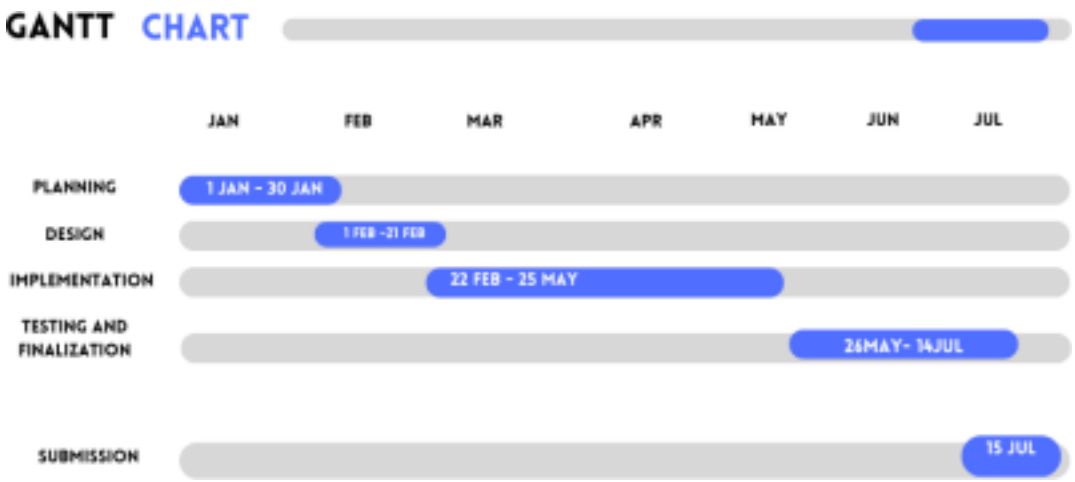
laptop	**Total**
Internet and hosting	
Applications access	

3.5 Timeline

A clear timeline was developed to manage project phases from planning to final deployment.

Task Category	Activity	Start Date	End Date	Duration (Weeks)
Planning	Literature Review & Requirements Finalization	4 Jan	30 Jan	1
Design	Database & System Design (ERD)	3 Feb	21 Feb	1
Implementation	Environment Setup, Module Development (Patient, Appointment, Security, Reporting)	10 Feb	25 May	22
Testing & Finalization	Testing	7 May	14 Jul	26
Milestone	Final Prototype Submission —	15 Jul	15 Jul	

Gantt Chart Representation:



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Ajami, S., & Arab-Chadegani, R. (2013). Barriers toimplementingt Electronic Health Records (EHRs). *Materia socio-medica*,*25*(3), 213–215. https://doi.org/10.5455/msm.2013.25.213-215

CHAPTER FOUR: IMPLEMENTATION, TESTING AND RESULTS

4.1 Implementation Approach

CuraTrack adopted the Prototyping Model as its implementation methodology. The iterative approach led to continuous refinement of the modules through repeated tests and validations. The system was developed using an Object-Oriented and Modular approach. The clinical functions, such as Patient Registration and Medical Records, were created as pure modules and then combined into the main Flask app.

Architectural Pattern: The system utilises the Model-View-Controller (MVC) design pattern. The (Model) part is MySQL database logic, while the (View) part is user interface programming. (Controller) Is the bridge written in Python Flask?

Rather than developing their own custom server logic, many projects reused the Flask Framework, thereby enabling powerful routing and sessions. The Jinja2 template was reused to populate the HTML table with patient data dynamically.

The clinical visit database must use relational database logic so that each clinical visit is always linked to a Patient ID. So even if there are many records, they won't be separated by a unique patient's correct ID.

4.2 Choice of Programming Languages and Tools.

The choice of the "Tech Stack" was driven by affordability and quick prototyping.

Python was chosen for its clean syntax and libraries for connecting to MySQL.

I have used MySQL as the RDBMS to reliably and securely manage structured health records.

Flask was chosen for its lightweight simplicity, in contrast to most larger frameworks. The aim here was to make it easy to maintain the system, even by personnel with minimal technical training.

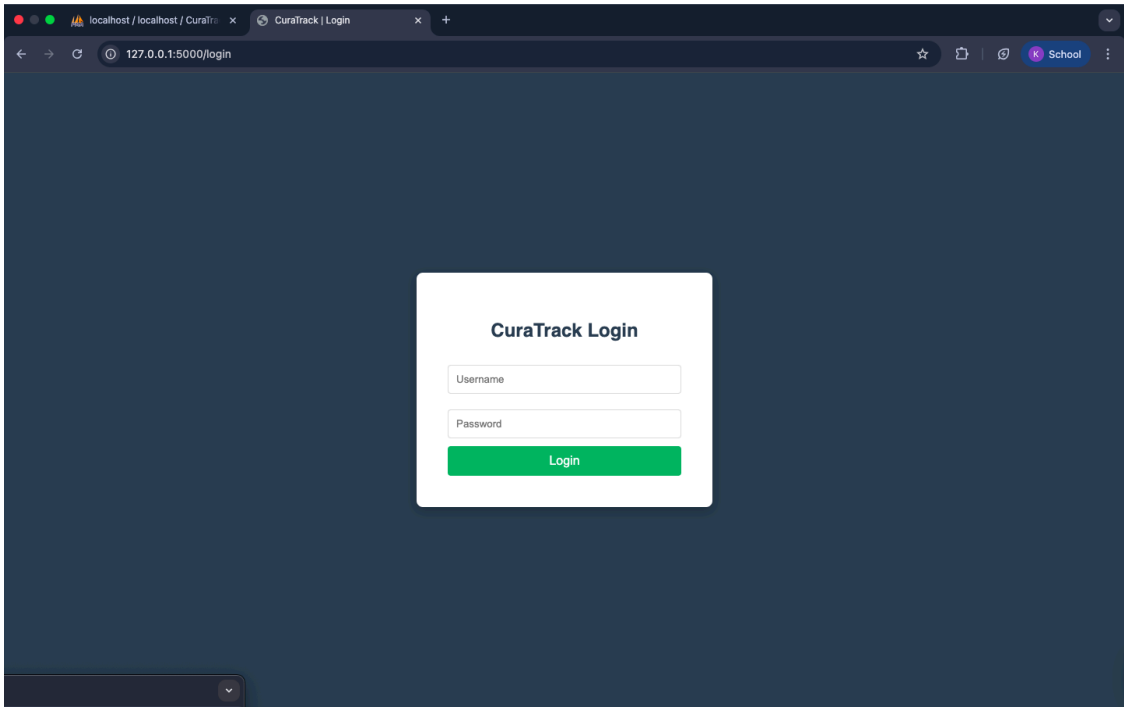
In this project, VS Code is used as the IDE for coding and debugging the Python/Flask logic.

4.3 Detailed System Operation (How it Works)

CuraTrack operates in a secure, connected workflow.

4.3.1 Authentication and Access Control

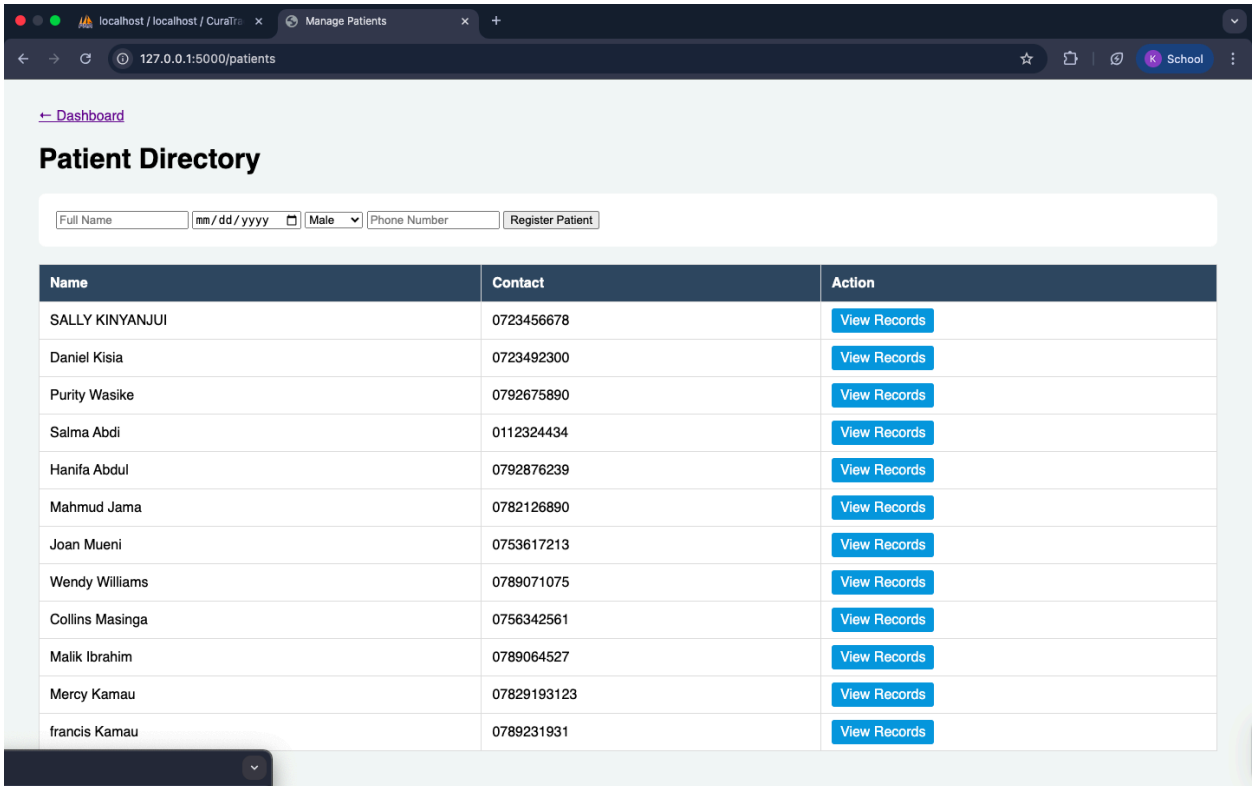
You can log on to the system only through the login page/module. One that allows only authorised users, provided via Role-Based Access Control (RBAC), so that the login administrator can manage system users and the provider (doctor/nurse) can access patient clinical data. '.



[Figure 4.1: Login Screen showing the authentication interface]

4.3.2 Patient Management and Directory

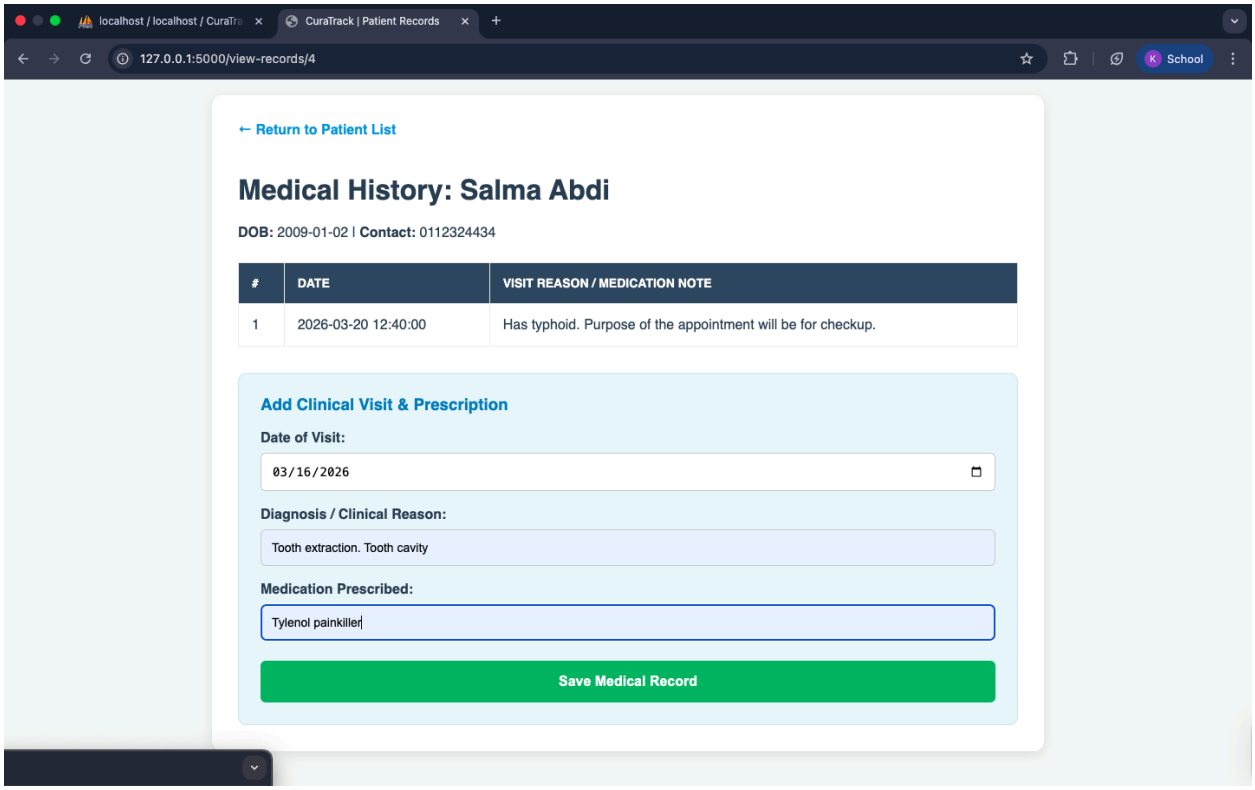
To access and enter a patient's medical record, the patient must first be registered in the patient management module, or the patient must be retrieved from the directory. When the user enters the new patient registration, demographic details, such as name, DOB, and contact information, must be captured.



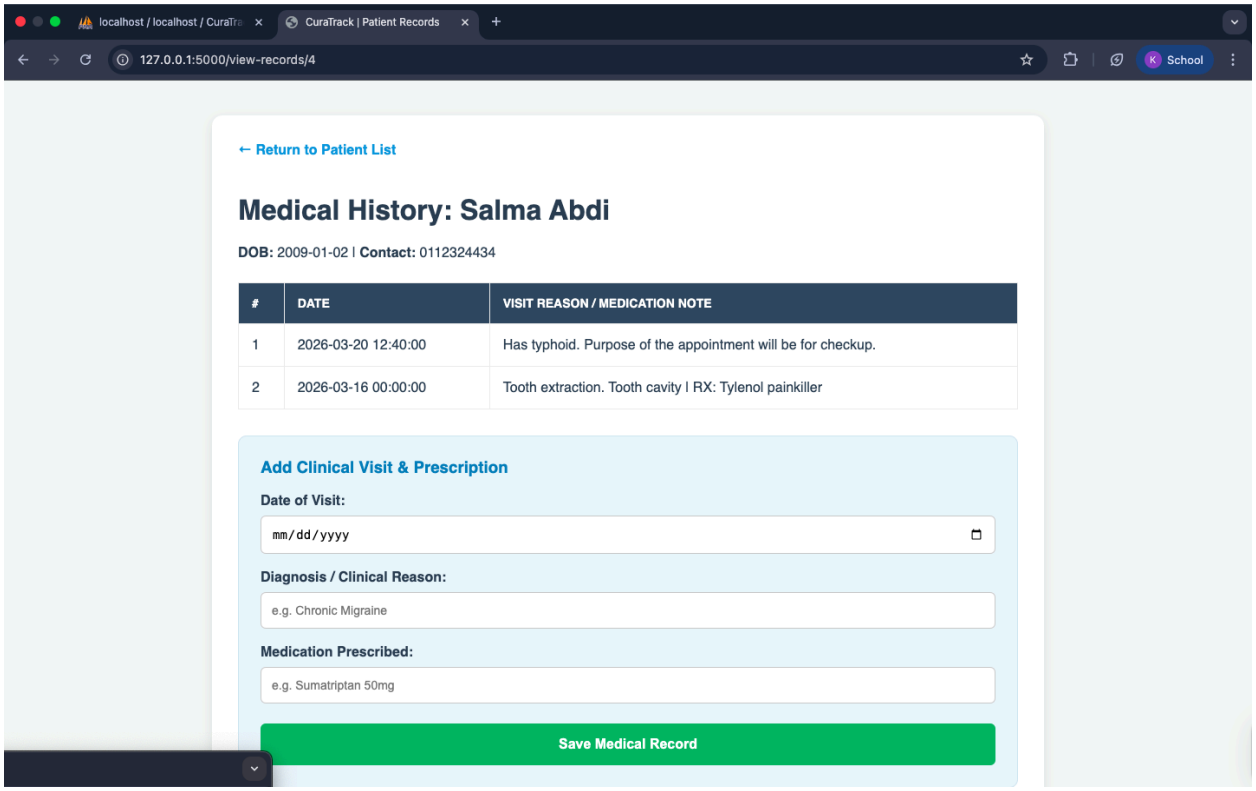
[Figure 4.2: Patient Directory showing a list of registered patients]

4.3.3 Clinical Record Entry and Prescription (Rx)

This is the system's main recording. In the record section, the vendor writes "Diagnosis" and "Medication Prescribed." For small clinics, for a quick overview of the patient's treatment history, these are concatenated in one chronological record, e.g. High Fever – Paracetamol to administer



[Figure 4.3: Medical History page showing a clinical visit being saved]



This is the evidence that it was actually saved.

4.4 System Testing and Verification

The system was verified through a structured Test Plan:

The Test Plan detailed below has verified the system:

- Switch Testing. It was conducted tests on every Python. Function, for instance, is adding a patient to confirm it. Receives data and discards empty data values.

System Testing. Combining the register patient testing was done on the module and the “Medical Records” module. This means that whenever a patient is added, they appear in the “Medical Records” module for entering clinical visits. UAT: The system underwent testing to mimic a standard simulation. Clinic Day. Add a patient, search for a patient, and add a prescription. For interactive use, the system must be intuitive.

The accomplished implementation resulted in a working prototype. This prototype meets the key project objectives outlined in the project.

The system can retrieve a patient's records in seconds via an SQL query, compared to the minutes it takes in the day-to-day manual search through paper files.

The system behind the above operation was automated to prevent errors such as duplicate data and documents with unreadable handwriting, since a paper-based system cannot be automated.

The system can be used on a laptop, tablet, or smartphone, as it’s web-based. As a result, the system can easily be transported.

CHAPTER FIVE: CONCLUSION, EVALUATION, AND FURTHER WORK

5.1 Introduction

The chapter offers an evaluation of the CuraTrack for executing this project. It provides insight into how well the project's objectives have been achieved, how the development process unfolded, and further research that can be carried out. The chapter concludes with a summary of the project's contribution to digital health management for small-scale healthcare providers.

The goal of this research was to develop a digital solution to the data loss and inefficient manual health record management system. The researcher achieved the above by meeting the objectives outlined in Chapter One.

Objective 1: Develop an integrated central database.

This was achieved: The MySQL relational database is a single source of truth. Normalization will thus minimise data duplication. For example, patient demographics are stored separately from the clinical visit log but linked to it via relational keys.

Objective 2: Develop an application interface for clinical data that is easy to use. Managed to develop an interface using the Flask framework and Jinja2 templating. Using the dashboard, patient directory, or medical history module, they can access data entry with minimal clicks. This solution aims to protect user data or provide alternative options for data hosting. The CuraTrack application uses transaction control to ensure database integrity and data consistency while handling multiple transactions. The system requires user authentication. Only those with the correct authorisation can access medical data. As a web application, it is usable on devices connected to the clinic’s local area network.

5.3 Critical Evaluation of the System

CuraTrack is a major improvement to paper filing.

The “Quick Action” buttons on the dashboard, along with searching for the patient’s name in the patient directory, help instantly retrieve the patient's medical history, reducing waiting time.

Data Integrity: Taking the example of a mandatory “Medication (Rx)” field in the clinical entry form. This ensures that a diagnosis isn’t stored without an entry for a treatment plan. Clinicians will have a more robust audit trail.

Many EMRs (especially the enterprise solution) tend to complicate things. The CuraTrack system is designed to be operated by staff with basic computer literacy.

5.4 Challenges Faced

The development journey encountered many technical and logistical difficulties.

Challenges faced while building that dashboard are:

- Maintain a relationship with the patient, and his multiple appointments will require “foreign-key” constraints within the database. This is to ensure they do not create “orphaned” records.
- Implementing dynamic updates for “Total Patients” and “Total Visits” on the dashboard required using correct aggregate SQL queries in the relevant Flask route.
- The issue of a logged-out user accessing our app via the back button was solved by a logout function that cleared the session, which was thoroughly tested.

5.5 Further Work and Recommendations

Integrate the system with a finance module so that, in the future, it will automatically generate invoices for clinical visits based on the prescribed medication.

Data visualisation is to be carried out to capture the trends in disease outbreaks or patient demographics over time.

A script that backs up the MySQL database to the cloud (AWS, Google Drive, etc.) whenever there is a failure in the hard drive.

5.6 Conclusion

The project allows us to conclude that cheap, open-source software, such as Python and MySQL, can indeed be used to solve real-world, existing problems. The patient's complete life cycle is covered within the system, from registration to prescription. The entire clinic process is stored in a computer rather than filed manually. CuraTrack has already been implemented at a Kenyan

clinic, enabling the digitisation of healthcare facilities. The project demonstrates how technology can transform service delivery with accuracy and speed, even with limited resources.